File permissions in Linux

Project description

In this project, I managed file and directory authorization for the `researcher2` user's `projects` directory in a Linux environment. As a security analyst, I was tasked with auditing and modifying file permissions to ensure compliance with organizational security policies that prohibit unauthorized write access. Through systematic permission analysis and modification using `chmod` commands, I restricted access to sensitive files, secured archived data, and protected directory contents from unauthorized group access.

Check file and directory details

cd projects
`ls -l`: To list permission excluding hidden files
`ls -la`: To list permission include hidden files

Current file permissions (from `ls -l  and  ls -la output`):

```
researcher2@a2f10fc46ad2:~$ cd projects
researcher2@a2f10fc46ad2:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jun 16 08:23 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Jun 16 08:23 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Jun 16 08:23 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jun 16 08:23 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jun 16 08:23 project_t.txt
researcher2@a2f10fc46ad2:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 16 08:23 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 16 08:36 ..
-rw--w---- 1 researcher2 research_team   46 Jun 16 08:23 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 16 08:23 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Jun 16 08:23 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Jun 16 08:23 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jun 16 08:23 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jun 16 08:23 project_t.txt
researcher2@a2f10fc46ad2:~/projects$
```

Describe the permissions string

Using `project_k.txt` as an example with permissions `-rw-rw-rw-`:
The 10-character string indicates file type and permission levels for three owner categories.
The 1st character (`-`) identifies this as a regular file (directories show `d`).

Characters 2-4 (`rw-`) represent user/owner permissions: read and write access.
Characters 5-7 (`rw-`) represent group permissions: read and write access.
Characters 8-10 (`rw-`) represent other/everyone permissions: read and write access.
When a permission shows a hyphen (`-`) instead of `r`, `w`, or `x`, that specific permission is not granted.

Change file permissions

File requiring modification: `project_k.txt`
Issue: This file had write permissions for "other" users (`-rw-rw-rw-`), violating the organizational policy that prohibits others from having write access to any files.

Command used: `chmod o-w project_k.txt`

```
researcher2@a2f10fc46ad2:~/projects$ chmod o-w project_k.txt
researcher2@a2f10fc46ad2:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Jun 16 08:23 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Jun 16 08:23 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Jun 16 08:23 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jun 16 08:23 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jun 16 08:23 project_t.txt
```

Output/Result:
After running the command and verifying with ls -l, the permissions changed from `-rw-rw-rw-` to `-rw-rw-r--`, successfully removing write access for other users while maintaining read-only access.

Change file permissions on a hidden file

File: `.project_x.txt` (hidden archived file)
Requirement: The user and group should be able to read the file, but no one should have write permissions since it's archived data.

Command used: `chmod u-w,g-w,g+r .project_x.txt`

```
researcher2@14613677184b:~/projects$ chmod u-w,g-w,g+r .project_x.txt
researcher2@14613677184b:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 16 09:11 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 16 09:30 ..
-r--r----- 1 researcher2 research_team   46 Jun 16 09:11 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 16 09:11 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Jun 16 09:11 project_k.txt
-rw------- 1 researcher2 research_team   46 Jun 16 09:11 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jun 16 09:11 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jun 16 09:11 project_t.txt
researcher2@14613677184b:~/projects$
```

Output/Result:
This command removed write permissions from both the user (`u-w`) and group (`g-w`) while ensuring the group maintained read access (`g+r`). The file permissions were updated to `-r--r-----`, preventing any modifications to the archived data while preserving read access for authorized users.

Change directory permissions

Directory: `drafts`
Requirement: Only the `researcher2` user should have access to this directory and its contents, which means only the owner should have execute privileges on the directory.
Command used: `chmod g-x drafts`

```
researcher2@14613677184b:~/projects$ chmod g-x drafts
researcher2@14613677184b:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 16 09:11 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 16 09:30 ..
-r--r----- 1 researcher2 research_team   46 Jun 16 09:11 .project_x.txt
drwx------ 2 researcher2 research_team 4096 Jun 16 09:11 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Jun 16 09:11 project_k.txt
-rw------- 1 researcher2 research_team   46 Jun 16 09:11 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jun 16 09:11 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Jun 16 09:11 project_t.txt
researcher2@14613677184b:~/projects$
```

Output/Result:
After running the command and verifying with `ls -la`, the directory permissions changed from `drwx--x---` to `drwx------`, successfully removing execute permission from the group. This ensures that only `researcher2` can access (`cd` into) the drafts directory and view its contents.

Summary

Through this authorization management exercise, I systematically audited and secured file permissions across the projects directory to enforce the principle of least privilege. By using `ls -la` to identify permission vulnerabilities and `chmod` to remove unauthorized write access, restrict archived files, and protect directory contents, I ensured compliance with organizational security policies. These tasks demonstrate essential Linux security skills including permission string interpretation, hidden file management, and access control implementation critical competencies for protecting sensitive data in cybersecurity operations.